

Tandberg TDV-2200 Complete Escape Sequence Analysis

Comprehensive Escape Sequence Table

Based on analysis of the TDV-2200 firmware and the ND terminal specification, here's the complete mapping of escape sequences.

Character Processing Overview

The TDV-2200 uses a sophisticated character processing system in `nd_character_processor` (0x1BE3):

```
assembly

; Main character dispatcher
CPI 0xFC      ; Check for ND escape trigger (252)
CZ 0x19CE     ; Call nd_escape_sequence_dispatcher if FC

; Extended parameter mode (207)
CPI 0xCF      ; Check for extended parameter mode (207)
JNZ next1
CALL 0x0CC1   ; Read multiple characters for complex sequences
JC 0x1BF6     ; Loop until complete

; Parameter mode (206)
CPI 0xCE      ; Check for parameter mode (206)
JNZ next2
CALL 0x1F69   ; Parse CSI parameters

; Handle ND private control sequences
CPI 0xB8      ; Special control (184)
CPI 0xB1      ; Control sequence 1 (177)
CPI 0xB2      ; Control sequence 2 (178)
CPI 0xB3      ; Control sequence 3 (179)
CPI 0xB4      ; Control sequence 4 (180)
```

Standard VT100 Escape Sequences

Sequence	Hex	VT100	TDV	Function	Implementation
ESC D	0x44	✓	✓	IND - Index (scroll down)	Standard VT100
ESC E	0x45	✓	✓	NEL - Next Line (CR+LF)	Standard VT100
ESC H	0x48	✓	✓	HTS - Horizontal Tab Set	Standard VT100
ESC M	0x4D	✓	✓	RI - Reverse Index (scroll up)	Standard VT100
ESC [	0x5B	✓	✓	CSI - Control Sequence Introducer	Extended in TDV
ESC c	0x63	✓	✓	RIS - Reset to Initial State	Standard VT100

ND/TDV Private Escape Sequences

Sequence	Hex	VT100	TDV	Function	Implementation
ESC 1	0x31	✗	✓	NDSS1 - Single shift charset 1	TDV-specific
ESC 2	0x32	✗	✓	NDSS2 - Single shift charset 2	TDV-specific
ESC 3	0x33	✗	✓	NDSS3 - Single shift charset 3	TDV-specific
ESC 4	0x34	✗	✓	NDSS4 - Single shift charset 4	TDV-specific
ESC 5	0x35	✗	✓	NDSS5 - Single shift charset 5	TDV-specific
ESC 6	0x36	✗	✓	NDSS6 - Single shift charset 6	TDV-specific
ESC 7	0x37	✓	✓	NDSC - Save Cursor	Similar to VT100 DECSC
ESC 8	0x38	✓	✓	NDRC - Restore Cursor	Similar to VT100 DECRC
ESC 9	0x39	✗	✓	NDSS7 - Single shift charset 7	TDV-specific
ESC :	0x3A	✗	✓	NDSS8 - Single shift charset 8	TDV-specific
ESC ;	0x3B	✗	✓	NDSS9 - Single shift charset 9	TDV-specific
ESC =	0x3D	✓	✓	NDEAKM - Enter Alt Keypad Mode	Similar to VT100 DECKPAM
ESC >	0x3E	✓	✓	NDXAKM - Exit Alt Keypad Mode	Similar to VT100 DECKPNM

TDV Private Control Codes

Code	Hex	VT100	TDV	Function	Implementation
252	0xFC	✗	✓	ND_ESC_TRIGGER	Triggers escape sequence processing
206	0xCE	✗	✓	ND_PARAM_MODE	Parameter parsing mode
207	0xCF	✗	✓	ND_EXT_PARAM_MODE	Extended parameter parsing
177	0xB1	✗	✓	ND_CTRL_SEQ_1	Control sequence 1
178	0xB2	✗	✓	ND_CTRL_SEQ_2	Control sequence 2
179	0xB3	✗	✓	ND_CTRL_SEQ_3	Control sequence 3
180	0xB4	✗	✓	ND_CTRL_SEQ_4	Control sequence 4
184	0xB8	✗	✓	ND_SPECIAL_CTRL	Special control function

TDV Private CSI Sequences (ESC [n _)

Work Area and Line Operations

Sequence	Decimal	Hex	VT100	TDV	Function	Implementation
ESC [112 p	112	0x70	✗	✓	NDLIWA - Insert lines in work area	FUN_ram_0e70
ESC [113 q	113	0x71	✗	✓	NDDLWA - Delete lines in work area	Not found
ESC [114 r	114	0x72	✗	✓	NDSTBM - Set top/bottom margin	Not found
ESC [115 s	115	0x73	✗	✓	NDICHE - Insert chars with extent	Not found
ESC [116 t	116	0x74	✗	✓	NDDCHE - Delete chars with extent	Not found

Rectangle Operations

Sequence	Decimal	Hex	VT100	TDV	Function	Implementation
ESC [117 u	117	0x75	✗	✓	NDSREC - Save rectangle	Not found
ESC [118 v	118	0x76	✗	✓	NDRREC - Restore rectangle	Not found
ESC [122 z	122	0x7A	✗	✓	NDSAR - Set attribute in rectangle	Not found
ESC [123 {	123	0x7B	✗	✓	NDAAR - Add attribute in rectangle	Not found
ESC [124	124	0x7C	✗	✓	NDRAR - Remove attribute in rectangle	Not found
ESC [125 }	125	0x7D	✗	✓	NDFC - Fill character in rectangle	Not found

System and Configuration

Sequence	Decimal	Hex	VT100	TDV	Function	Implementation
ESC [119 w	119	0x77	✗	✓	NDSSKL - Set soft-key level	Not found
ESC [120 x	120	0x78	✗	✓	NDREQ - Request terminal parameters	Not found
ESC [121 y	121	0x79	✗	✓	NDTST - Invoke confidence test	Links to diagnostics
ESC [126 ~	126	0x7E	✗	✓	NDWA - Define work area	Not found
ESC [127 DEL	127	0x7F	✗	✓	NDVIDEO - Alpha video on/off	Not found

Additional TDV CSI Sequences Found

Lower Range (0x01-0x6F)

Range	Count	VT100	TDV	Function	Implementation
ESC [1-9 _	9	✗	✓	Unknown TDV functions	Stored in memory
ESC [10-69 _	60	✗	✓	Unknown TDV functions	Stored in memory
ESC [70-87 _	18	✗	✓	ND private extensions	Some implemented

Function Implementation Analysis

Character Processing Entry Point

assembly

```
; nd_character_processor (0x1BE3)
; Handles all incoming characters and escape sequences
```

```
CPI 0xFC      ; ND escape trigger
CZ 0x19CE     ; -> nd_escape_sequence_dispatcher
```

```
CPI 0xCF      ; Extended parameter mode
CALL 0x0CC1   ; -> try_read_serial_data (multi-char sequences)
```

```
CPI 0xCE      ; Parameter mode
CALL 0x1F69   ; -> nd_parse_csi_parameters
```

```
; ND private control sequences
CPI 0xB1-0xB4 ; Control sequences 1-4
CPI 0xB8      ; Special control
```

Escape Sequence Dispatcher

assembly

```
; nd_escape_sequence_dispatcher (0x19CE)
; Executes escape sequences via function pointer
```

```
LHLD (0x5E08) ; Load nd_escape_handler_function_pointer
MOV A,H       ; Check if pointer is valid
ORA L
RZ            ; Return if NULL pointer
PCHL         ; Jump to escape handler function
```

CSI Parameter Parser

assembly

```
; nd_parse_csi_parameters (0x1F69)
; Parses numeric parameters for CSI sequences

CALL 0x1F85      ; -> nd_validate_parameter_digit (first digit)
JC error         ; Exit on invalid digit
RLC              ; Shift left 4 bits (high nibble)
RLC
RLC
RLC
MOV B,A          ; Save high nibble

CALL 0x1F85      ; -> nd_validate_parameter_digit (second digit)
JC error
ANI 0x0F         ; Mask low nibble
ORA B            ; Combine with high nibble
```

Comments Added to Disassembly

assembly

```
; Address 0x1BE3 - nd_character_processor
; ND Terminal Character Processor: Handles ND private escape sequences
; and CSI commands according to ND specification
; 0xFC=Escape trigger, 0xCE/0xCF=Parameter parsing, 0xB1-0xB4=Control sequences

; Address 0x19CE - nd_escape_sequence_dispatcher
; ND Escape Sequence Handler: Executes escape sequence via function pointer
; at 0x5E08. Supports ND private sequences like NDSS1-NDSS9, NDSC, NDRC, CSI sequences

; Address 0x1F69 - nd_parse_csi_parameters
; ND CSI Parameter Parser: Parses decimal parameters for CSI sequences
; Format: ESC [ nnn _ where nnn is 3-digit decimal parameter

; Address 0x1F85 - nd_validate_parameter_digit
; ND Parameter Digit Validator: Validates digits 0-9 and hex A-F for parameters

; Address 0x0E70 - TDV private function for CSI 112 p (NDLIWA)
; Insert lines in work area - handles line insertion with extent parameters
```

Summary

VT100 Compatibility:

- **Standard sequences:** ESC D, E, H, M, [, c, 7, 8, =, >

- **Compatible but extended:** CSI sequences enhanced with TDV features

TDV-Specific Features:

- **9 alternative character sets** (NDSS1-NDSS9)
- **87 private CSI sequences** (ESC [1-87 _)
- **Rectangle operations** (save/restore/manipulate screen areas)
- **Work area control** (insert/delete lines and characters)
- **Soft key programming** (NDSSKL)
- **Built-in diagnostics** (NDTST confidence tests)
- **Custom control codes** (0xB1-0xB4, 0xB8, 0xCE, 0xCF, 0xFC)

Implementation Status:

- **Character processing:** Fully implemented with sophisticated dispatch system
- **Parameter parsing:** Complete CSI parameter parsing system
- **Function dispatch:** Dynamic function pointer system for escape handlers
- **Specific sequences:** Only a few specific CSI functions found (like 0x70/112p)
- **Most CSI sequences:** Stored in memory but implementation functions not identified

The TDV-2200 is a **highly advanced terminal** that significantly extends VT100 capabilities with professional features for sophisticated applications requiring advanced screen manipulation, multiple character sets, and work area control.